Jai Sachin Baskar

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RESEARCH INTERESTS

Human Computer Interaction, Usable Privacy and Security, Responsible AI, Online Safety.

EDUCATION

SRM Institute of Science and Technology, India | B.Tech in Computer Science and Engineering — June 2022 – May 2026
GPA: 8.65/10 | Bachelor's Thesis: 10/10

Indian Institute of Technology Madras | Foundational Level, Programming and Data Science (32 Credits) — Jan 2023 – Sept 2024

RESEARCH EXPERIENCE

Indian Institute of Technology Gandhinagar — May 2026 – Present
Research Assistant, FUSS Group (Formal and Usable Systems Security)
Supervisors: Dr. Abhishek Bichhawat and Dr. Mainack Mondal (IIT Kharagpur)

- Engineered parallel, cross-platform NLP pipelines analyzing 1M+ reviews split across Android and iOS marketplaces, mapping localized threat vocabularies (e.g., "chor app") and regional code-switching to capture safety signals that standard Western models exclude.
- Developed a data-stratification framework that clustered the 1M+ reviews into 4 distinct linguistic groups (Standard English, misspellings, Indian geographic nuances, and Hinglish) to systematically analyze underrepresented Global South user populations.
- Spearheaded a mixed-methods user research study analyzing why genuine user reviews are deleted from the Google Play Store, comparing deletion patterns against developer-declared privacy labels to map user security concerns.

The University of Queensland, Australia — Apr 2025 – Sep 2025
Research Assistant | Behavioural Cybersecurity
Supervisor: Dr. Sam Macaulay

- Conducted expert cognitive walkthroughs using the Awareness, Motivation, and Capability (AMC) framework to evaluate attacker decision-making and operational ROI against advanced hardware countermeasures like decoy logic and power randomization.
- Executed behavioral threat modeling on anti-tamper integrated circuits, identifying critical user-bypass vulnerabilities such as executing chip delayering inside a vacuum to neutralize air-reactive, pyrophoric security mechanisms.
- Analyzed the economic feasibility of reverse-engineering (RE) workflows by benchmarking adversary capabilities against multi-million dollar RE tooling, reframing hardware robustness through a psychological and cost-asymmetric usable security lens.

Indian Institute of Information Technology Kottayam, Kerala — May 2023 – Nov 2023
Research Intern, Cybersecurity Group

- Designed a logic-locking system grounded in a core usable security principle: security mechanisms should impose asymmetric friction, making legitimate use seamless while raising the effort cost for attackers disproportionately.
- Implemented UART-based key authentication so legitimate users interact through a familiar, low-friction interface while the hardware actively resists key extraction attempts from untrusted foundries.
- Solo-authored the resulting paper, awarded **Best Paper at IEEE ICCCIT 2025**, and filed a design patent (No. 202541088733, 2025).

HEC Paris, France — Sep 2024 – Dec 2024
Research Assistant
Supervisor: Dr. Yasir Dewan

- Discovered critical failure modes in standard BERT-based models when classifying a dataset of 12,000 strategic and behavioral records, pivoting to an in-depth qualitative study to extract domain-specific keywords and linguistic nuances.
- Transformed the qualitative linguistic findings into a custom algorithmic Python pipeline, improving classification accuracy to 85% across the 12,000 records by successfully capturing context that standard transformer models missed.

University of Sydney, Australia — Jan 2024 – Jul 2024
Research Assistant
Supervisor: Dr. Vijaya Murthy

- Conducted a comprehensive systematic literature study the socio-technical trust and accountability implications of automated auditing, identifying critical vulnerabilities in human-AI collaboration and systemic data governance.

SRM Research Lab, Chennai — Jan 2023 – Apr 2026
Student Researcher

- Developed a computationally efficient deep learning model for automated intracranial haemorrhage detection in emergency CT scans, designed for real-time integration into clinical workflows **(IEEE ICOEI 2026; Bachelor's Thesis, 10/10)**.
- Built DeepRoadGuard, a multi-sensor Bayesian fusion model that improves pothole detection accuracy in low-visibility and night-time driving conditions **(IEEE ICCDS 2025)**.

- Modelled consumer purchase behaviour using data mining and predictive analytics to identify decision drivers and optimise retail inventory strategy **(arXiv 2024; 2nd Best Research Paper Award)**.

PUBLICATIONS

- Jaisachin B., "Enhancing Security in Logic-Locked Systems through UART-Based Key Authentication," Proc. IEEE ICCCIT, pp. 649-653, 2025. **[Best Paper Award; Solo-Authored]**
- Jaisachin B., "Computationally Efficient Deep Learning for Automated Intracranial Haemorrhage Detection in Emergency CT Scans," Proc. IEEE ICOEI, 2026. **[First Author; Bachelor's Thesis, 10/10]**
- Akshay G. N., Jaisachin B., "DeepRoadGuard: Enhanced Pothole Detection Using Bayesian Fusion and Night-Vision Data," Proc. IEEE ICCDS, 2025.
- Sri Darshan M., Jaisachin B., "Integrating Data Mining and Predictive Modelling Techniques for Enhanced Retail Optimisation," arXiv:2409.19248, 2024. **[Best Paper Award; Runner-Up]**

PATENT

- Jaisachin B., *UART-Based Key Authentication in a Logic-Locked System.* Patent No. 202541088733, Filed 2025.

HONOURS AND AWARDS

Dean's and Merit Scholarship | New York University, Tandon School of Engineering	2026
Indian Delegate | HPAIR Conference (Harvard Project for Asian and International Relations)	2025
Best Research Paper Award | Preventing IP Theft in Integrated Circuits, IEEE (Winner)	2024
Best Research Paper Award | Data Mining and Predictive Modelling for Retail Optimisation (Runner-Up)	2024
Runner-up, Web Development Hackathon | SRM University	2024
Runner-up, Machine Learning Hackathon | Epoch, SRM University	2023
Gold Medalist | National Computer Science Olympiad	2018

ACADEMIC SERVICE AND MEMBERSHIPS

- IEEE Student Member; IEEE Electron Devices Society Member

LEADERSHIP AND ACTIVITIES

Research and Development Lead | Cyborg Club, SRM Institute	2023 – 2025
Public Relations Lead | Dhyan Chand Sports Club, IIT Madras	2023 – 2024
Social Science Teacher, Volunteer | Bihar, India	2024
Member | Ramanujam Society of Research, IIT Madras	2023 – 2025
Student Volunteer | National Service Scheme, India	2023

SKILLS

HCI and Usable Security: User Studies, Thematic Analysis, Survey Design, Behavioural Data Analysis, Human Factors in Security, Privacy Perception Research, Mixed-Methods Design

NLP and Data Science: Natural Language Processing, Large-Scale Text Mining, Data Mining, Machine Learning, Bayesian Fusion, Power BI

Programming and Tools: Python (NumPy, Pandas, Matplotlib, BeautifulSoup), C, SQL

Security and Hardware: Reverse Engineering, Logic Locking, IC Security, Vitis HLS, Vivado HLS

Languages: English, Hindi, Tamil (native trilingual), Malayalam (novice)